Research of Key Technologies for Automatic Acceptance of Electric Power E-commerce Materials Based on Image and Data-driven Approaches

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Abstract

Various emerging technologies and concepts are constantly being integrated into the power system to improve the efficiency of power grid operation. The types of power equipment and materials involved in multiple links such as power generation, energy storage, transmission, distribution and power consumption are increasing, which brings great challenges to the comprehensive management of power e-commerce. Especially in the acceptance stage of materials, the traditional manual acceptance method is inefficient and easily affected by human factors. This study aims to develop an automatic acceptance system for power e-commerce materials based on image recognition and data-driven to improve the intelligence level, accuracy and efficiency of the material acceptance process. By combining image recognition technology, GPS location information and weight sensor data, a new automatic acceptance method is proposed. The system uses sliding window filtering and error removal algorithms to preprocess data, and generates symmetric key encryption to transmit images to ensure the security of data transmission. Furthermore, the CNN model is used to review the matching degree of acceptance materials to ensure that the accepted materials are the items required by the purchase order. The experimental results show that the system can achieve a high correct acceptance rate under various working conditions, effectively reducing the cost of manual participation and improving the intelligence level of the power material management process.

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CCS Concepts

• **Information systems** → Information systems applications; Process control systems.

Keywords

Electric Power E-commerce, Automatic Acceptance, Image Recognition, Data-driven, CNN

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1 Introduction

With the improvement of the digitalization and intelligence level of power grids, the application of emerging technologies in power systems is increasing [1]. Against the backdrop of the ubiquitous development of power networks and the intelligent development of power management, the introduction of new technologies such as network communication equipment, intelligent management facilities, renewable energy generation, energy storage systems, and electric vehicle charging infrastructure has brought about many new types of equipment. The development of clean energy [2] generation technologies such as wind power and solar power has expanded the power input sources of traditional power grids, while also increasing the types and complexity of power grid equipment.

In the first generation of power generation systems represented by thermal power generation [3], the equipment is relatively simple, generally with large generators as the core, the distribution network is relatively simple, the transmission lines are mainly medium and low voltage, and the management and operation of power materials are relatively simple. With the incorporation of hydropower generation [4] into the power grid, the entry of nuclear power generation [5] into the starting stage, and the interconnection and integration of power networks, automated control systems have begun to appear to improve the stability and safety of power networks.

Long-distance high-voltage transmission technology has also begun to be used, the types of equipment and materials in the power grid have become increasingly complex, and material management has also begun to be improved through informatization. With the advancement of technology, ultra-high voltage technology [6] has been gradually applied, smart grids [7] have also been gradually formed, smart devices in the power grid have begun to be applied, and distribution automation equipment has gradually increased. While improving the efficiency and automation level of the power grid, it has further increased the difficulty of material management. The development of wind and solar clean energy [8] power generation technology has also further increased the types and quantities of materials that need to be managed and maintained in the power grid, and has raised the requirements for material management. In addition to power generation and transmission, electricity demand is also undergoing profound changes. For example, the popularization of electric vehicles requires the deployment of a large number of charging piles during urban construction [9], which will also bring certain challenges to the layout of the power network and the management of materials and equipment.

Against the backdrop of the rapid transformation and development of the power industry and the rapid increase in the complexity of the technical system, the procurement and management of power materials are also facing profound changes. The existing power e-commerce platform has been able to realize automatic and intelligent centralized procurement of equipment and materials, but it still relies on traditional manual operations during acceptance, which has problems such as low efficiency, poor accuracy, high cost, and greater influence of human factors. In order to further improve the intelligent level of power material management, the automatic acceptance of power e-commerce materials has become a hot topic in current research and application.

In order to further improve the efficiency and intelligence level of material management methods, this study aims to develop an automatic acceptance system for power e-commerce materials based on image recognition and data-driven to improve the efficiency and accuracy of material acceptance, reduce the cost of manual participation, and improve the intelligence level of the power supply chain. Through image-based intelligent recognition, the classification and counting of materials and equipment are realized, and the digital information collected by sensors such as position and weight is further combined to make a comprehensive evaluation of material acceptance. At the same time, environmental parameters are used for encrypted transmission of data information, effectively improving the security and reliability of the system.

2 Materials and Methods

The automatic acceptance process of materials mainly includes the steps of material arrival detection, data collection and preprocessing, data analysis and acceptance judgment, acceptance result generation and feedback, and exception handling. The system collects image information of materials through multi-angle cameras, collects acceptance location information through GPS sensors, and collects weight information through standard scales. Among them, the location information and weight information will be preprocessed to generate a symmetric key, and the collected image will be encrypted and transmitted. The server will decrypt and process the goods image according to the weight of the goods in the purchase order and the scheduled delivery location data, so as to improve the security of the system and avoid the leakage of goods information.

2.1 Overall Design

In the material procurement stage, the purchase order describes the purchaser's information and the category of materials. The system can obtain the weight, category and other information of the materials to be delivered, as well as data such as the delivery address based on the purchase order. Among them, the confirmation of the delivery address can not only prevent the materials from being handed over to the wrong warehouse, but also prevent the occurrence of false delivery. Video image information is the most important basis for judgment in the automatic acceptance of materials. Image recognition is used to determine whether the materials to be accepted are the types of materials defined in the purchase order to ensure that the materials delivered are the materials needed by the purchaser. Confirmation of characteristic information such as the weight of materials is a further guarantee of the authenticity of the materials, which is conducive to preventing the materials with the same appearance but simple internal modules from being inferior.

Therefore, data information such as video images, location and weight are all important data information for the automatic acceptance system to drive the acceptance process. Packaging all this data information with asymmetric keys and sending them to the acceptance server is a potential method, but the amount of video image data is large, and the encryption and decryption process will reduce the response time of the system. A special feature of this system is that the acceptance server already knows the delivery location and weight of the materials in advance based on the purchase order, so this information can be confirmed in a special way. Since different materials differ in terms of delivery location and characteristics, we can use this information as a data source for generating symmetric keys. On the one hand, encryption and decryption using symmetric keys can improve system speed and reduce system load. On the other hand, the acceptance terminal and the server can only decrypt video image data when the encryption and decryption keys are consistent, which is another form of confirmation of the consistency of the acceptance location and material attributes.

Figure 1 shows the overall process of material procurement after the integration of the material automatic acceptance system, which mainly includes the following steps: 1. Users submit material procurement applications from the centralized procurement platform according to production business needs. 2. The management personnel of the centralized procurement platform publish bidding information based on the application and organize interested units to participate in the bidding. After a company successfully bids, a procurement list is generated. 3. The centralized procurement platform sends the purchase order information to the automatic acceptance server, so that the acceptance server can obtain the acceptance location (GPS information as the basis for judgment), material type (video image recognition as the basis for judgment) and material attribute information (weight and other information

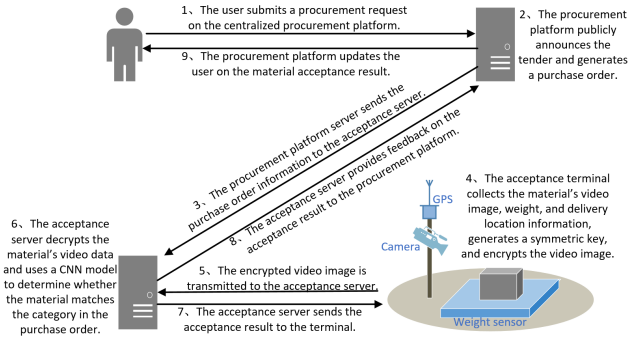

Figure 1: Schematic diagram of material procurement and automatic acceptance process.

as the basis for judgment) of the materials to be accepted. 4. After the materials arrive, the automatic acceptance terminal collects the weight and video image information of the materials, the GPS information of the delivery location, generates a symmetric key through GPS and weight, and encrypts the video image. 4. The acceptance terminal sends the encrypted image to the acceptance server. 5. The acceptance server generates a symmetric key based on the information in the purchase order to decrypt the video image, and then identifies whether the materials to be accepted are consistent with the purchase category based on the CNN model. 6. The acceptance server feeds back the acceptance conclusion to the acceptance terminal. 7. After the acceptance is passed, the acceptance server feeds back the acceptance information to the centralized procurement platform. 8. The centralized procurement platform pushes the arrival information of materials to users.

2.2 Data Collection and Key Generation

Once the materials arrive, the system automatically collects the image, weight, and location data and preprocesses them to ensure accuracy and consistency. Due to fluctuation errors in GPS positioning data (x, y, z) and weight data w, preprocessing using a sliding window method [10] is required to filter out large error data. First, the sliding window average is calculated for the x, y, and z components of the location data.

$$x_{\text{avg}} = \frac{1}{N} \sum_{i=k-N}^{k-1} x_i \tag{1}$$

$$y_{\text{avg}} = \frac{1}{N} \sum_{i=k-N}^{k-1} y_i \tag{2}$$

$$z_{\text{avg}} = \frac{1}{N} \sum_{i=k-N}^{k-1} z_i \tag{3}$$

The distance between the current position $(x_k, y_k, z_k)$and the sliding window average $(x_{avg}, y_{avg}, z_{avg})$ can be calculated as:

$$d_k = \sqrt{(x_k - x_{\text{avg}})^2 + (y_k - y_{\text{avg}})^2 + (z_k - z_{\text{avg}})^2} \tag{4}$$

If the distance d_k exceeds a predetermined threshold d_{th} , the current point is considered an outlier and discarded. Similarly, a sliding window is applied to the weight data to eliminate outliers:

$$w_{\text{avg}} = \frac{1}{N} \sum_{i=k-N}^{k-1} w_i \tag{5}$$

When the absolute difference between the current weight value w_k and the sliding window average w_{avg} exceeds a threshold W_{th}, the weight data is considered an outlier and discarded:

$$(\left|w_k - w_{\text{avg}}\right| > W_{th}) \tag{6}$$

To mitigate the effects of manufacturing tolerances and delivery strategy deviations, the data is further quantized:

$$\varphi_{quantized} = \frac{\varphi}{\Delta\varphi} \times \Delta\varphi, \quad \varphi = x, y, z, \ or \ w \tag{7}$$

Based on the weight $w_{quantized}$ and the processed location $(x_{quantized}, y_{quantized}, z_{quantized})$, a symmetric key can be generated using a hash function [11]:

$$K = hash\left(x_{quantized}, y_{quantized}, z_{quantized}, w_{quantized}\right) \tag{8}$$

Compared with asymmetric encryption, the video images sent by the acceptance terminal to the server are encrypted based on the generated dynamic key, and the server can decrypt faster. The generated symmetric keys vary depending on the type of material, ensuring the dynamic and reliability of the key.

2.3 Image Collection and Processing

After the acceptance end collects the video image of the materials to be accepted, it is encrypted with the key K and further sent to the remote server for acceptance judgment. Only when the keys match can the server correctly decrypt the image and compare it with the feature vector of the materials required in the purchase order to determine whether it can be approved for acceptance. The image processing process adopts the CNN [12] model, which consists of multiple convolutional layers, pooling layers, fully connected layers and activation functions. The convolution layer convolves the input image through the convolution kernel (or filter) to extract the local features of the image. Assuming the input image is I, the convolution kernel is K, and the output of the convolution operation is

$$O_{i,j} = (I * K)_{i,j} = \sum_{m=1}^{M} \sum_{n=1}^{N} I_{i+m-1,j+n-1} \cdot K_{m,n} \tag{9}$$

where $M \times N$ is the size of the convolution kernel. ReLU activation is applied to the output of the convolution layers. The pooling layers downsample the data to reduce dimensionality, with the output of a 2×2 max-pooling operation given by:

$$P_{i,j} = max\left\{O_{2i-1,2j-1}, O_{2i-1,2j}, O_{2i,2j-1}, O_{2i,2j}\right\} \tag{10}$$

Flatten $P_{i,j}$ into a vector x and input it to the fully connected layer:

$$y = Wx + b \tag{11}$$

where W and b are the weight matrix and bias term. Finally, a Softmax function converts the output into a probability distribution for classification:

$$P_j = \frac{exp\left(y_j\right)}{\sum_{k=1}^{C} exp\left(y_k\right)} \tag{12}$$

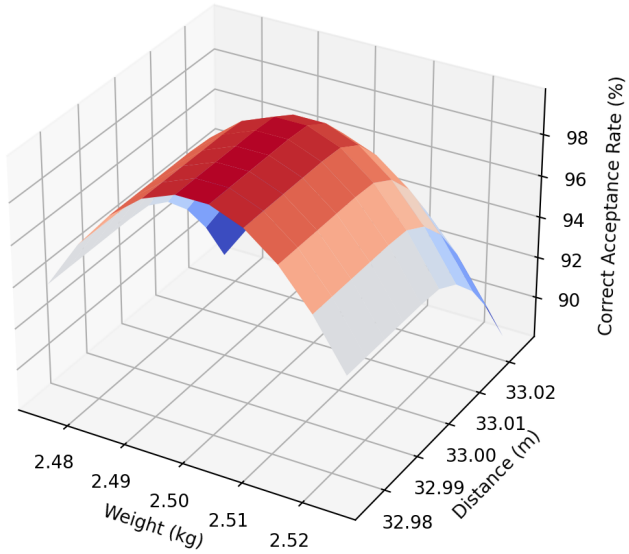

Figure 2: Probability of correct acceptance in different situations.

where C is the total number of classes, and y_j is the probability of class j. The trained model extracts high-level feature representations from the image, and these feature vectors are used for image matching. The similarity between a pre-trained feature vector f_1 and the real-time recognized feature vector f_2 is computed as:

$$\mathrm{S}\left(f_1, f_2\right) = \frac{f_1 \cdot f_2}{\| f_1 \| \| f_2 \|} \tag{13}$$

If the similarity $\mathrm{S}(f_1, f_2)$ exceeds a predetermined threshold S_{th}, the match is successful, and the system accepts the material. Otherwise, an anomaly notification is sent to the acceptance terminal. If the system cannot complete the automatic acceptance, manual intervention will investigate whether the material matches the procurement order.

3 Experimental Results

In order to test the effectiveness of the algorithm, we conducted an experimental verification of the automatic acceptance system. The material to be accepted is an electric energy meter with a weight of 2.5kg and a volume of 29cm×17cm×8.5cm. The distance threshold d_{th} and weight threshold W_{th} are set to 50m and 20g respectively. By adjusting the GPS antenna position of the acceptance site and the weight of the acceptance item, the probability that the system can complete the automatic acceptance is tested. The results are shown in the figure below:

Figure 2 shows the experimental results, in which we can see that at the standard weight and delivery location, the system achieves a maximum correct acceptance rate of 97.3%. As the distance increases from the standard position, the system maintains a high correct acceptance rate for distances less than 33m, but the acceptance rate rapidly decreases as the distance continues to grow. Due to inherent GPS positioning inaccuracies, the correct acceptance rate begins to decline even when the actual distance is slightly less than the threshold. The system cannot pass acceptance for materials inspected at distant, non-target locations. Although there may be some manufacturing deviations in the material's weight, setting a reasonable W_{th} ensures effective acceptance accuracy.

4 Conclusions

In this study, we designed and implemented an automatic acceptance system for electric power e-commerce materials based on image recognition and sensor data. By incorporating weight, GPS location, and multi-source data, the system significantly improves the accuracy and automation of the acceptance process. Through error filtering and sliding window methods, the system ensures reliability in real-world applications. Experimental results show that the system can achieve a high rate of automatic acceptance under various conditions. This solution reduces labor costs and enhances the intelligence and efficiency of the power material management process, offering important insights for managing materials on power e-commerce platforms. Future work will focus on applying multimodal methods in automatic acceptance systems to integrate diverse data and improve material acceptance accuracy and reliability.

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